Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, SQL, Java, C/C++, R
Data Analytics & ETL: Data Gathering, Data Cleaning, Data Normalization, ETL Pipelines, Exploratory Data Analysis (EDA), Feature Engineering, Statistical Analysis
Databases & Data Engineering: SQLite, MySQL, MongoDB, Redis, DBMS Fundamentals, Relational Database Design, Data Modeling, Query Optimization
Data Visualization & BI: Power BI, DAX, Tableau, Matplotlib, Plotly, Dashboard Development, Data Reporting
Machine Learning & AI: Scikit-learn, PyTorch, Anomaly Detection, Time-Series Analysis, LLMs, RAG, Prompt Engineering
Tools & Engineering Practices: Git/GitHub, FastAPI, Flask, React, PyTest, Test-Driven Development (TDD), Jupyter Notebook

EXPERIENCE

VOIS – Vodafone Idea Foundation **|LINK** — ***Aug – Sep, 2025***

- Analysed 2 real-world business datasets containing 8,500+ records, applying data preprocessing and AI-assisted analytics to identify trends and generate 15 actionable business insights for data-driven decision-making.
- Validated analytical findings for accuracy and consistency, reviewing outputs to ensure reliable and meaningful insights for business recommendations.
- Established a standardised analytics workflow covering data preprocessing, insight generation, and validation, enabling consistent conversion of raw business data into actionable recommendations.

PROJECTS

Financial Transaction Anomaly Detection & Audit Dashboard **| LINK**

- Engineered an end-to-end Python analytics pipeline processing 50K+ PaySim transactions, incorporating data validation, duplicate detection, statistical outlier analysis, vendor-velocity checks, HHI concentration analysis, and Isolation Forest.
- Designed a star-schema SQLite data warehouse and implemented a composite 0–100 risk-scoring framework, identifying 357 high/critical-risk transactions with end-to-end analysis completed in under 7 seconds.
- Developed a 5-page Power BI dashboard with 20+ DAX measures to deliver vendor risk, anomaly-rate, and transaction trend insights, supported by automated pipeline tests.

Customer Lifetime Value Prediction & Segmentation **|LINK**

- Processed and analyzed 1M+ retail transactions to identify purchasing patterns, evaluate customer behavior, and uncover customer retention and revenue opportunities.
- Developed BG/NBD and Gamma-Gamma models for customer purchase and monetary value estimation, achieving an R^2 of 0.87 for 12-month customer value prediction.
- Built interactive Power BI dashboards and an ROI simulator to translate customer segments and revenue forecasts into actionable retention and business insights.

AI-Powered Expense Receipt Scanner & Analytics Platform **|LINK**

- Developed an automated receipt-processing pipeline using EasyOCR and OpenCV to extract expense data from receipt images, supported by a FastAPI backend and React frontend.
- Implemented SQLAlchemy-based data persistence with SQLite to organize and manage extracted expense records in a structured relational database.
- Integrated JWT authentication and Google OAuth for secure access, and deployed the application using Render and Vercel, reducing processing time by 40%.

Real-Time Stock Market Anomaly Detector **|LINK**

- Engineered a Python-based analytics pipeline processing 2M+ NSE market records, combining price signals and financial-news sentiment to identify unusual market behavior.
- Developed and evaluated Isolation Forest and LSTM Autoencoder models, achieving 88% precision and 85% recall on validation data for market anomaly detection.
- Optimized the analytics pipeline to reduce detection latency by 35% and developed interactive Streamlit and Plotly visualizations for monitoring anomalies and analyzing market trends.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** — 2026
- Deloitte Australia Data Analytics Job Simulation |**LINK** — 2025